Data Storytelling:
Introduction to Text as Data

Patrick Chester
Credit: Arthur Spirling
Week 11

April 6, 2026

Announcements

- Homework 1 is graded! I'll be uploading scores with comments on Canvas right after class.
- Areas for improvement:
  - If you are having difficulty rendering documents in quarto, this is a big issue that needs to be addressed. I cannot grade what I cannot see.
  - Explore data thoroughly. Many people lost points when they failed to see the -9 missing values in the age variable
  - Read all parts of each question. If a question asks to explain a result, please do so. Many points were lost when you were asked to explain the correlation results.
    - What does a correlation of 0.24 between age and leave represent? What direction and strength does it indicate between these variables?
- Homework 2 will be distributed tonight or early tomorrow. It will cover data visualization, data collection, and text analysis. Perhaps some geospatial analysis as well!

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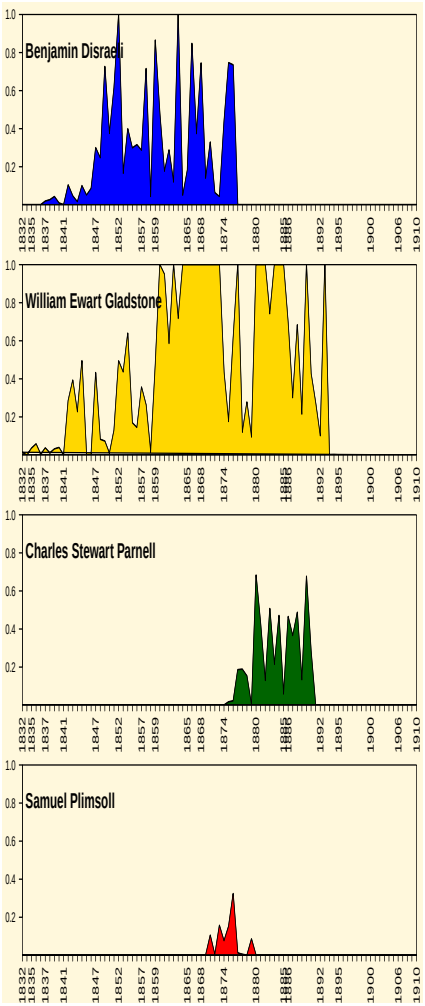

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but may not be as important as you think.

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e.g. stemmed word like ‘`treasuri`’, which doesn’t appear in the document itself.

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e.g. "`Brown vs Board of Education`" may not be usefully tokenized as '`Brown`', '`vs`', '`Board`', '`of`', '`Education`'

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NB these words mean something ‘special’ (and slightly opaque) when combined. Related to idea of collocations: words that appear together more often than we’d predict based on random sampling.

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e.g. working with Congressional speech data, '`representative`' might be a stop word; in *Hansard* data, '`honourable`' might be.

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a	about	above	after	again	against	all
am	an	and	any	are	aren't	as
at	be	because	been	before	being	below
between	both	but	by	can't	cannot	could
couldn't	did	didn't	do	does	doesn't	doing
don't	down	during	each	few	for	from
further	had	hadn't	has	hasn't	have	haven't
having	he	he'd	he'll	he's	her	here
here's	hers	herself	him	himself	his	how
how's	i	i'd	i'll	i'm	i've	if
in	into	is	isn't	it	it's	its
itself	let's	me	more	most	mustn't	my
myself	no	nor	not	of	off	on
once	only	or	other	ought	our	ours
ourselves	out	over	own	same	shan't	she
she'd	she'll	she's	should	shouldn't	so	some
such	than	that	that's	the	their	theirs
them	themselves	then	there	there's	these	they
they'd	they'll	they're	they've	this	those	through
to	too	under	until	up	very	was
wasn't	we	we'd	we'll	we're	we've	were
weren't	what	what's	when	when's	where	where's
which	while	who	who's	whom	why	why's
with	won't	would	wouldn't	you	you'd	you'll
you're	you've	your	yours	yourself	yourselves	

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→ annotating in this way is called parts-of-speech tagging.

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In practice, need something faster (and cruder), so software implements the Porter Stemmer using algorithms like `Snowball`.

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Original Word		Stemmed Word
abolish	$\mapsto$	`abolish`
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1 `The mountains are beautiful in Ore. and Wash.`

2 `http://www.wsj.com/articles/son-of-saul-not-about-the-survivors-1449590175`

3 `I can't go with him to Beijing.`

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also can use *substrings* which are groups of *n* contiguous characters.

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a military patrol boat rescued three of the kayakers on general carrera
lake and a helicopter lifted out the other three the chilean army said
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"general carrera" "carrera lake" "lake and" "and a" "a helicopter"
"helicopter lifted" "lifted out" "out the" "the other" "other three"
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lake" "carrera lake and" "lake and a" "and a helicopter" "a helicopter
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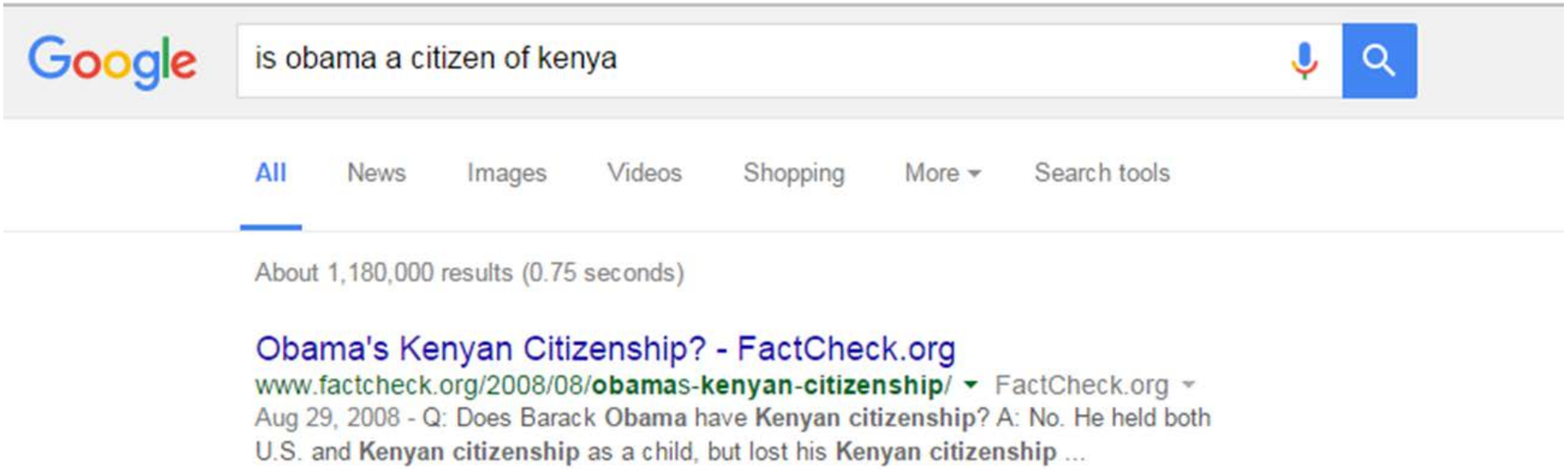

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Obama's Kenyan Citizenship? - FactCheck.org
www.factcheck.org/2008/08/**obamas-kenyan-citizenship**/ FactCheck.org
Aug 29, 2008 - Q: Does Barack **Obama** have **Kenyan citizenship**? A: No. He held both U.S. and **Kenyan citizenship** as a child, but lost his **Kenyan citizenship** ...

Google

"is obama a citizen of kenya?"

All News Images Videos Shopping More Search tools

3 results (0.37 seconds)

Is Obama a citizen of Kenya? - Blockland Forum
forum.blockland.us/index.php?topic=77191.25;imode
Is Obama a citizen of Kenya? << < (6/8) > >>. Garuda: Ah yes, the conservative agenda, full of "My country tis of thee" bullshit. In the way that they always blast ...

Is Obama really a U.S. citizen | Woodstock Sentinel Review
www.woodstocksentinelreview.com/.../is-oba... Woodstock Sentinel-Review

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```
                 features
docs                 american expect induct presid will
  1933-Roosevelt            2      1      1      1   12
  1937-Roosevelt            4      0      0      2   16
  1941-Roosevelt            4      0      0      1    4
  1945-Roosevelt            1      0      0      1    7
```

partial TDM from Roosevelt's Inaugural Addresses

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```
           docs
features   1933-Roosevelt 1937-Roosevelt 1941-Roosevelt 1945-Roosevelt
  american               2               4               4               1
  expect                 1               0               0               0
  induct                 1               0               0               0
  presid                 1               2               1               1
  will                  12              16               4               7
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Notes on a DTM

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NB there are efficient ways to store and manipulate sparse matrices.

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Is a generative model insofar as it seeks to model the joint distribution between class and features (cf 'discriminative' models that only care about $Pr(\text{class}|x)$)

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$$\Pr(A|B) \propto \Pr(A)\Pr(B|A)$$

Here, $\Pr(A)$ is our prior for A, while $\Pr(B|A)$ will be the likelihood for the data we saw.

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3. A subject claims to have psychic abilities—he can tell you how a (fair) coin will come down in nine tosses. He has less than a $\frac{1}{500}$ chance of being correct by chance, but he succeeds in the task! Do you 'update' that he has psychic abilities? Why or why not?

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Jihadi Clerics

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Can assign a *Jihad Score* to each document: basically the logged likelihood ratio, $\sum_i \log \frac{\Pr(t_k|\text{Jihad})}{\Pr(t_k|\neg\text{ Jihad})}$ (note: doesn't know what 'real world' priors are, so drops them here)

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Then for each cleric, concatenate all works into one and give this 'document'/cleric a score.

Discriminating Words

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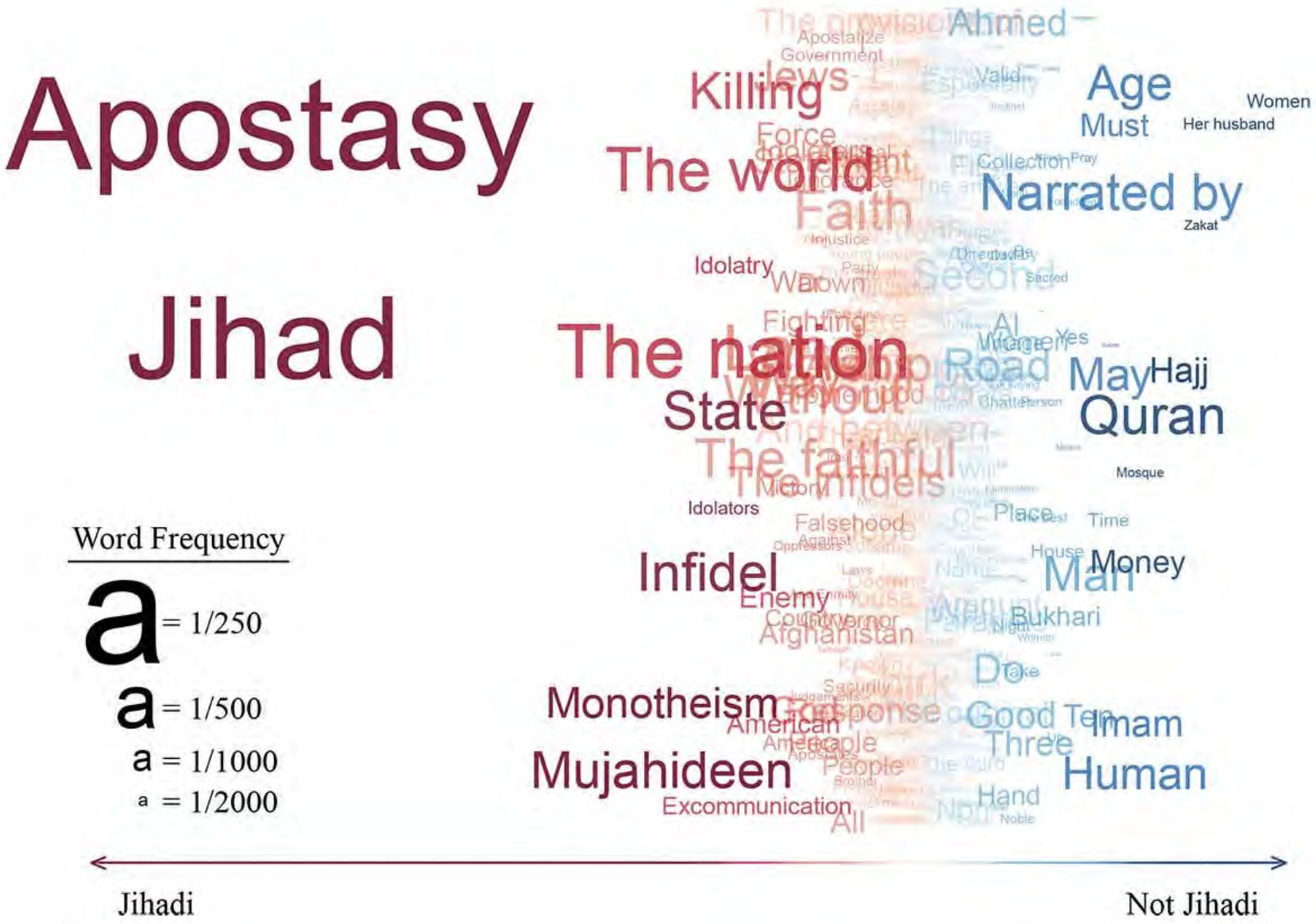

Validation: *Exoneration*

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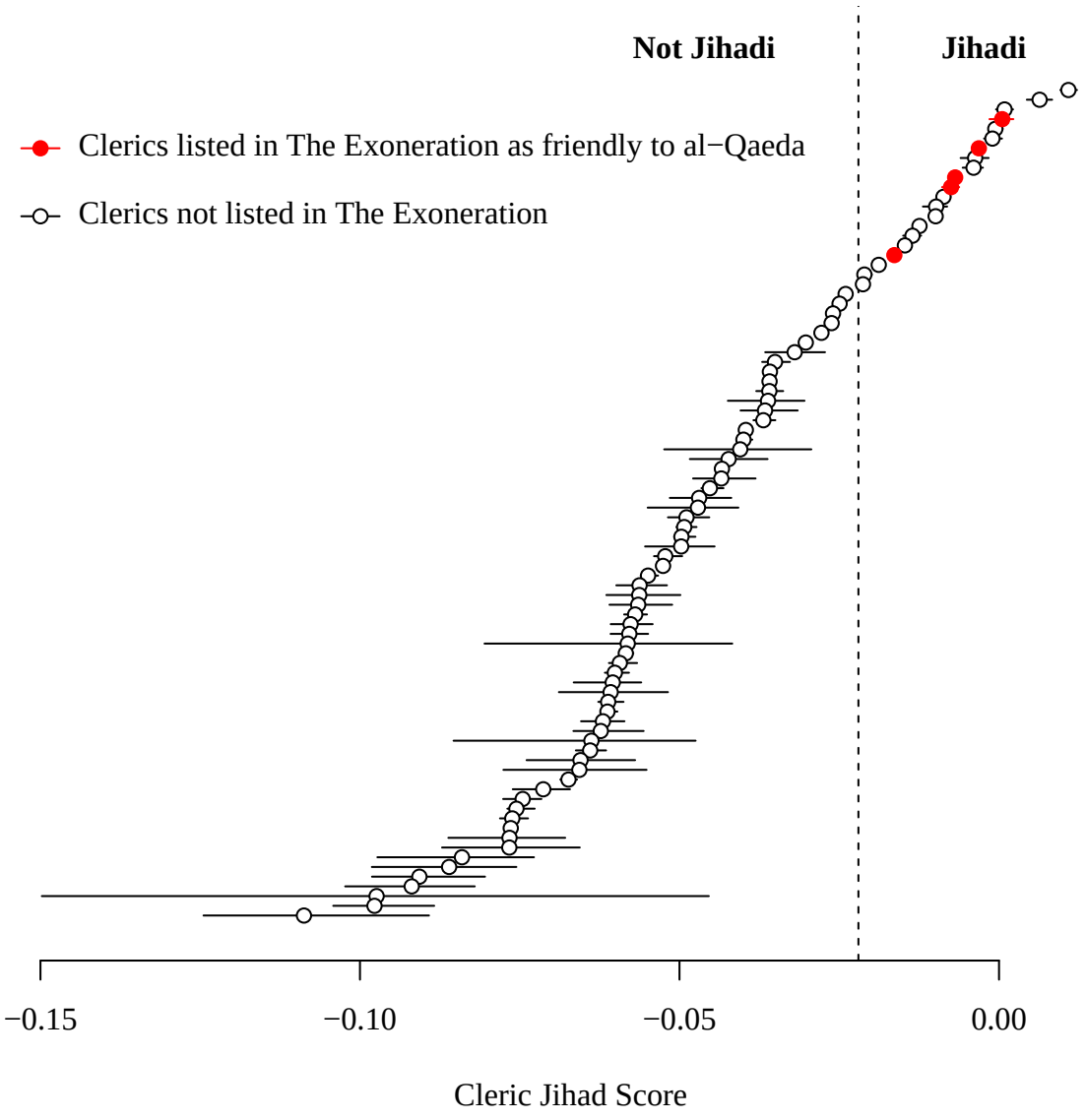

Figure 4.9: *Jihad Scores Predict Inclusion in The Exoneration*

That's it! Now let's take a look at the code for today!